

FCAI ANU Materials – Semi Seniors – Chapter 5 – Visual Programming

Chapter 5: Modularizing Your Code with Methods - Quiz

1. What is a primary benefit of using methods?

- a) They make the program run faster
- b) They break a complex program into smaller, manageable pieces
- c) They automatically fix syntax errors
- d) They are only used for event handling

Answer: b) They break a complex program into smaller, manageable pieces

2. A void method is one that:

- a) Returns a value to the statement that called it
- b) Always requires arguments
- c) Executes a group of statements and then terminates without returning a value
- d) Must be declared as public

Answer: c) Executes a group of statements and then terminates without returning a value

3. Which of the following is a special type of method that responds to events?

- a) Value-returning method
- b) Event handler
- c) Constructor
- d) Main method

Answer: b) Event handler

4. A program that is broken into smaller units of code, such as methods, is known as a:

- a) Procedural program
- b) Object-oriented program
- c) Modularized program
- d) Structured program

Answer: c) Modularized program

5. The benefit of writing a method once and using it multiple times is known as:

- a) Code simplification
- b) Code reuse
- c) Code optimization
- d) Code documentation

Answer: b) Code reuse

6. What are the two main parts of a method definition?

- a) Name and body
- b) Header and footer
- c) Header and body
- d) Access modifier and return type

Answer: c) Header and body

7. If a method is declared as private, it can be called:

- a) By code in any class
- b) Only by code inside the same class
- c) Only by event handlers
- d) By code in a different namespace

Answer: b) Only by code inside the same class

8. In a method header, what does the keyword void indicate?

- a) The method is empty
- b) The method returns a value
- c) The method does not return a value
- d) The method accepts no arguments

Answer: c) The method does not return a value

9. Which naming convention is typically used for method names in C#?

- a) camelCase
- b) Pascal Case
- c) snake_case
- d) kebab-case

Answer: b) Pascal Case

10. The method header is terminated with a:

- a) Semicolon (;)
- b) Colon (:)
- c) Comma (,)
- d) It is not terminated with a semicolon

Answer: d) It is not terminated with a semicolon

11. The statements performed when a method is executed are contained in the:

- a) Method header
- b) Method body
- c) Parameter list
- d) Access modifier

Answer: b) Method body

12. Where must method definitions be written?

- a) Outside of any class
- b) Inside a namespace but outside a class
- c) Inside the class to which they belong
- d) In a separate file

Answer: c) Inside the class to which they belong

13. How do you call a method named DisplayMessage?

- a) call DisplayMessage;
- b) DisplayMessage;
- c) DisplayMessage();
- d) void DisplayMessage();

Answer: c) DisplayMessage();

14. In the Lights application example, what does the switchButton_Click method do?

- a) It only turns the light on
- b) It only turns the light off
- c) It reverses the state of the light by calling TurnLightOn or TurnLightOff
- d) It closes the application

Answer: c) It reverses the state of the light by calling TurnLightOn or TurnLightOff

15. A piece of data passed into a method when it is called is known as a(n):

- a) Parameter
- b) Argument
- c) Modifier
- d) Expression

Answer: b) Argument

16. A variable that receives an argument passed into a method is known as a(n):

- a) Parameter
- b) Attribute
- c) Instance
- d) Field

Answer: a) Parameter

17. Consider the method: private void DisplayValue(int value). What is int value inside the parentheses?

- a) An argument
- b) A return type
- c) A parameter variable declaration
- d) An access modifier

Answer: c) A parameter variable declaration

18. Which of the following is a valid call to a method that accepts an integer argument?

- a) `DisplayValue("5");`
- b) `DisplayValue(int 5);`
- c) `DisplayValue(5);`
- d) `Call DisplayValue(5);`

Answer: c) `DisplayValue(5);`

19. In the Argument Demo application, what does the `demo3Button_Click` method do?

- a) Calls `DisplayValue` once with the value 5
- b) Calls `DisplayValue` with the result of `3+5`
- c) Uses a loop to call `DisplayValue` five times with values 0 through 4
- d) Passes a string to the `DisplayValue` method

Answer: c) Uses a loop to call `DisplayValue` five times with values 0 through 4

20. In the Cards application, the `ShowCard` method accepts what type of parameter?

- a) `int`
- b) `PictureBox`
- c) `string`
- d) `bool`

Answer: c) `string`

21. When passing multiple arguments to a method, the arguments are matched to parameters based on their:

- a) Names
- b) Data types
- c) Positions
- d) Memory addresses

Answer: c) Positions

22. What is wrong with the following method header: `private void ShowMax(int num1, num2)`?

- a) The access modifier should be public
- b) The return type is missing
- c) A data type for num2 is missing
- d) The method name is invalid

Answer: c) A data type for num2 is missing

23. What is the feature that allows you to specify which parameter an argument should be passed into by using the format `parameterName: value`?

- a) Positional arguments
- b) Default arguments
- c) Named arguments
- d) Value arguments

Answer: c) Named arguments

24. Given the method `private void ShowName(string firstName, string lastName)`, which call uses named arguments correctly?

- a) `ShowName("Smith", "Suzanne");`
- b) `ShowName(firstName="Suzanne", lastName="Smith");`
- c) `ShowName(lastName: "Smith", firstName: "Suzanne");`
- d) `ShowName("Suzanne", lastName: "Smith");`

Answer: c) `ShowName(lastName: "Smith", firstName: "Suzanne");`

25. What is a default argument?

- a) An argument that must be provided by the caller
- b) A value that is automatically used for a parameter if no argument is provided
- c) An argument that is always zero
- d) The first argument in a parameter list

Answer: b) A value that is automatically used for a parameter if no argument is provided

26. In the method `private void ShowTax(decimal price, decimal taxRate = 0.07m)`, what is the default value for `taxRate`?

- a) 0.0m
- b) 0.05m
- c) 0.07m
- d) 1.0m

Answer: c) 0.07m

27. Which call to the `ShowTax` method is valid, given its definition with a default argument for `taxRate`?

- a) `ShowTax();`
- b) `ShowTax(100.0m);`
- c) `ShowTax(, 0.08m);`
- d) `ShowTax(0.08m, 100.0m);`

Answer: b) `ShowTax(100.0m);`

28. When an argument is passed by value, what happens?

- a) The method can change the original argument's value
- b) A reference to the argument is passed
- c) Only a copy of the argument's value is passed
- d) The argument must be a constant

Answer: c) Only a copy of the argument's value is passed

29. In the Pass By Value example, after the `ChangeMe` method executes, what is the value of the variable `number` in the `goButton_Click` method?

- a) 0
- b) 99
- c) It is undefined
- d) It depends on the user input

Answer: b) 99

30. What is the purpose of passing an argument by reference?

- a) To prevent the method from changing the argument's value
- b) To allow the method to change the argument's value in the calling code
- c) To make the program run faster
- d) To reduce memory usage

Answer: b) To allow the method to change the argument's value in the calling code

31. Which keyword is used to declare a reference parameter in C#?

- a) out
- b) ref
- c) param
- d) in

Answer: b) ref

32. What is a requirement for a variable being passed as an argument to a ref parameter?

- a) It must be a constant
- b) It must be uninitialized
- c) It must already be set to some value
- d) It must be a string

Answer: c) It must already be set to some value

33. In the Pass By Ref application, after the SetToZero method is called on variables x, y, z, what are their values?

- a) They remain 99, 100, and 101
- b) They become 0, 0, and 0
- c) They become 1, 1, and 1
- d) The values are random

Answer: b) They become 0, 0, and 0

34. How does an output parameter differ from a reference parameter?

- a) An argument for an output parameter must be initialized first
- b) A method is not required to set an output parameter's value
- c) An argument for an output parameter does not have to be set to a value before it is passed
- d) The ref keyword is used for output parameters

Answer: c) An argument for an output parameter does not have to be set to a value before it is passed

35. Which keyword is used to declare an output parameter?

- a) ref
- b) out
- c) param
- d) output

Answer: b) out

36. What is a requirement for a method that has an output parameter?

- a) It must not change the parameter's value
- b) It must set the output parameter to a value before it finishes
- c) It must return a value other than void
- d) It must be declared as public

Answer: b) It must set the output parameter to a value before it finishes

37. In the North America application, the GetFileName method uses an output parameter to return what?

- a) A list of countries
- b) A StreamReader object
- c) The selected filename
- d) An error message

Answer: c) The selected filename

38. What is a value-returning method?

- a) A method that performs a task but returns no value
- b) A method that returns a value to the statement that called it
- c) A method that only contains a void return type
- d) A method that must be called from another class

Answer: b) A method that returns a value to the statement that called it

39. Which of the following is a value-returning method from the .NET Framework?

- a) MessageBox.Show
- b) Console.ReadLine
- c) int.Parse
- d) Form1.Close

Answer: c) int.Parse

40. What two key things are required when writing a value-returning method that are not required for a void method?

- a) An access modifier and a parameter list
- b) A specified return data type and a return statement
- c) A method body and a name
- d) The public keyword and multiple parameters

Answer: b) A specified return data type and a return statement

41. In the method `private int Sum(int num1, int num2)`, what is the return type?

- a) private
- b) int
- c) Sum
- d) (int num1, int num2)

Answer: b) int

42. What does the return statement do in a value-returning method?

- a) It ends the program
- b) It prints the value to the screen
- c) It causes the method to end and sends a value back to the caller
- d) It declares a new variable

Answer: c) It causes the method to end and sends a value back to the caller

43. In the Sum application, the Sum method is called from the calculateButton_Click event handler. What is passed to it?

- a) The text from the age1TextBox and age2TextBox
- b) The integer values stored in userAge and friendAge
- c) The labels themselves
- d) The combinedAge variable

Answer: b) The integer values stored in userAge and friendAge

44. A Boolean method returns a value of what type?

- a) int
- b) string
- c) bool (true or false)
- d) decimal

Answer: c) bool (true or false)

45. What is the purpose of the IsEven method in the example?

- a) To calculate the square of a number
- b) To determine if a number is positive
- c) To determine if a number is even
- d) To convert a number to a string

Answer: c) To determine if a number is even

46. In the Pay and Bonus application, the InputIsValid method uses what keyword for its parameters pay and bonus?

- a) void
- b) ref
- c) out
- d) params

Answer: b) ref

47. What is the role of the InputIsValid method?

- a) To calculate the retirement contribution
- b) To close the application
- c) To convert user input and return true if successful, false otherwise
- d) To display the results in a label

Answer: c) To convert user input and return true if successful, false otherwise

48. The InputIsValid method returns what data type?

- a) decimal
- b) string
- c) int
- d) bool

Answer: d) bool

49. In the calculateButton_Click method of the Pay and Bonus app, when is the contribution calculated and displayed?

- a) Always, regardless of input validity
- b) Only if the IsValid method returns true
- c) Only if the IsValid method returns false
- d) Before the IsValid method is called

Answer: b) Only if the IsValid method returns true

50. A method that is declared as public can be called:

- a) Only by code inside the same class
- b) By code that is outside the class
- c) Only by event handlers
- d) Only by methods in the same file

Answer: b) By code that is outside the class

51. Which part of a method header specifies the data type of the value that the method returns?

- a) Access modifier
- b) Method name
- c) Return type
- d) Parameter list

Answer: c) Return type

52. What is the correct way to call a method named CalculateTotal that returns a decimal and assign its result to a variable named total?

- a) CalculateTotal(total);
- b) total = CalculateTotal();
- c) Call CalculateTotal() to total;
- d) return CalculateTotal(total);

Answer: b) total = CalculateTotal();

53. In the Lights application, what is the initial state of the lightOffPictureBox.Visible property?

- a) true
- b) false
- c) It depends on user action
- d) null

Answer: b) false

54. What happens when a method finishes execution?

- a) The program terminates
- b) Control returns to the statement that called the method
- c) The method remains in memory
- d) All variables in the program are reset

Answer: b) Control returns to the statement that called the method

55. The process of breaking a problem into smaller pieces handled by separate methods is a demonstration of:

- a) Object-Oriented Programming
- b) Modularization
- c) Exception Handling
- d) Inheritance

Answer: b) Modularization

56. Which of the following is NOT a valid argument that can be passed to a method?

- a) A variable
- b) A literal value
- c) An expression
- d) A method header

Answer: d) A method header

57. The TryParse methods are examples of:

- a) Void methods
- b) Event handlers
- c) Value-returning methods
- d) Constructors

Answer: c) Value-returning methods

58. What is the key difference in how ref and out parameters handle argument initialization?

- a) ref requires the argument to be initialized; out does not
- b) out requires the argument to be initialized; ref does not
- c) Both require the argument to be initialized
- d) Neither requires the argument to be initialized

Answer: a) ref requires the argument to be initialized; out does not

59. In the context of methods, what does "passing by reference" mean?

- a) Passing a copy of the value
- b) Passing the memory address of the variable
- c) Passing the name of the variable
- d) Passing a constant value

Answer: b) Passing the memory address of the variable

60. The statement `this.Close();` in the `exitButton_Click` method is an example of:

- a) Calling a value-returning method
- b) Calling a void method
- c) Declaring a method
- d) Passing an argument by reference

Answer: b) Calling a void method

Wish you all the best!

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